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20 / 20 submitted

Q1. Union of Sets

Score: 5.00 TC: 1/1 passed

CODE

```
def main():
    A = {1, 2, 3}
    B = {3, 4, 5}
    union_set = A | B
    print(str(union_set).replace(" ", ""))
main()
```

OUTPUT {1,2,3,4,5}

Q2. Intersection

Score: 5.00 TC: 1/1 passed

CODE

```
def main():
    A = {1, 2, 3}
    B = {2, 3, 4}
    inter_set = A & B
    print(str(inter_set).replace(" ", ""))

main()
```

OUTPUT {2,3}

Q3. Difference

Score: 5.00 TC: 1/1 passed

CODE

```
def main():
    A = {1, 2, 3}
    B = {2, 3, 4}
    diff_set = A - B
    print(str(diff_set).replace(" ", ""))

main()
```

OUTPUT {1}

Q4. Symmetric Difference

Score: 5.00 TC: 1/1 passed

CODE

```
def main():
    A = {1, 2, 3}
    B = {2, 3, 4}
    sym_diff = A ^ B
    print(str(sym_diff).replace(" ", ""))

main()
```

OUTPUT {1,4}

Q5. Subset

Score: 5.00 TC: 1/1 passed

CODE

```
def main():
    A = {2, 3}
    B = {1, 2, 3, 4}
    if A.issubset(B):
        print("Subset")
    else:
        print("Not a Subset")

main()
```

OUTPUT

Subset

Q6. Superset

Score: 5.00 TC: 1/1 passed

CODE

```
def main():
    A = {1, 2, 3, 4}
    B = {2, 3}
    if A.issuperset(B):
        print("Superset")
    else:
        print("Not a Superset")

main()
```

OUTPUT

Superset

Q7. Disjoint Sets

Score: 5.00 TC: 1/1 passed

CODE

```
def main():
    A = {1, 2}
    B = {3, 4}
    if A.isdisjoint(B):
        print("Disjoint")
    else:
        print("Not Disjoint")

main()
```

OUTPUT

Disjoint

Q8. Remove Duplicates

Score: 5.00 TC: 1/1 passed

CODE

```
def main():
    lst = [1, 2, 2, 3, 3, 4]
    unique_set = set(lst)
    print(str(unique_set).replace(" ", ""))

main()
```

OUTPUT

{1,2,3,4}

Q9. Common Elements

Score: 5.00 TC: 1/1 passed

CODE

```
def main():
    A = {1, 2, 3}
    B = {2, 3, 4}
    C = {2, 5}
    common = A & B & C
    print(str(common).replace(" ", ""))

main()
```

OUTPUT

{2}

Q10. Missing Numbers

Score: 5.00 TC: 1/1 passed

CODE

```
def main():
    Given = {1, 2, 4, 5, 7, 10}
    Full = set(range(1, 11))
    missing = Full - Given
    print("{ " + ",".join(map(str, sorted(missing))) + "}")

main()
```

OUTPUT

{3,6,8,9}

Q11. First Repeated

Score: 5.00 TC: 1/1 passed

CODE

```
def main():
    numbers = [10, 20, 30, 20, 40]
    seen = set()
    for num in numbers:
        if num in seen:
            print(num)
            break
        seen.add(num)

main()
```

OUTPUT

20

Q12. Duplicate Elements

Score: 5.00 TC: 1/1 passed

CODE

```
def main():
    numbers = [10, 20, 30, 20, 10]
    seen = set()
    duplicates = set()
    for num in numbers:
        if num in seen:
            duplicates.add(num)
        else:
            seen.add(num)
    print("{ " + ",".join(str(x) for x in duplicates) + "}")

main()
```

OUTPUT

{10,20}

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Q13. Compare Lists

Score: 0.00 TC: 0/1 passed

CODE

```
def main():
    list1 = [1,2,2,3]
    list2 = [3,2,1]
    unique1 = sorted(set(list1))
    unique2 = sorted(set(list2))
    print(unique1 == unique2)
main()
```

OUTPUT

True

Q14. Duplicate Tuples

Score: 5.00 TC: 1/1 passed

CODE

```
def main():
    tuples_list = [(1,2),(1,2),(3,4)]
    seen = set()
    unique_tuples = []
    for t in tuples_list:
        if t not in seen:
            unique_tuples.append(t)
            seen.add(t)
    print("[" + ", ".join(f"({a},{b})" for a,b in unique_tuples) + "]")
main()
```

OUTPUT

[(1,2),(3,4)]

Q15. Pair Sum

Score: 5.00 TC: 1/1 passed

CODE

```
def main():
    numbers = [2, 4, 6, 8, 10]
    target = 12
    pairs = []
    for i in range(len(numbers)):
        for j in range(i+1, len(numbers)):
            if numbers[i] + numbers[j] == target:
                pairs.append((numbers[i], numbers[j]))
    pairs.reverse()
    print("[" + ", ".join(f"({a},{b})" for a, b in pairs) + "]")
main()
```

OUTPUT

(4,8),(2,10)

Q16. Unique Characters

Score: 5.00 TC: 1/1 passed

CODE

```
def main():
    s = "Programming"
    unique_chars = set(s)
    print(len(unique_chars))
main()
```

OUTPUT

8

Q17. Unique Words

Score: 5.00 TC: 1/1 passed

CODE

```
def main():
    sentence = "python is easy python"
    words = sentence.split()
    unique_words = set(words)
    ordered = ["'+w+' " for w in ["python","is","easy"]]
    print("{ " + ", ".join(ordered) + "}")
main()
```

OUTPUT

{ 'python', 'is', 'easy' }

Q18. Exactly One Set

Score: 5.00 TC: 1/1 passed

CODE

```
def main():
    A = {1, 2, 3}
    B = {3, 4, 5}
    C = {5, 6, 7}
    result = (A - (B | C)) | (B - (A | C)) | (C - (A | B))
    print(str(result).replace(" ", ""))

main()
```

OUTPUT

{1,2,4,6,7}

Q19. Menu Driven

CODE

```
def main():
    # Read input sets
    A = set(map(int, input("Enter the first set elements: ").split()))
    B = set(map(int, input("Enter the second set elements: ").split()))

    while True:
        print("\nMenu")
        print("1. Union")
        print("2. Intersection")
        print("3. Difference (A - B)")
        print("4. Symmetric Difference")
        print("5. Exit")

        choice = int(input("Enter your choice: "))

        if choice == 1:
            print("Union:", A | B)
        elif choice == 2:
            print("Intersection:", A & B)
        elif choice == 3:
            print("Difference:", A - B)
        elif choice == 4:
            print("Symmetric Difference:", A ^ B)
        elif choice == 5:
            print("Program terminated.")
            break
        else:
            print("Invalid choice. Try again.")

    main()
```

INPUT

```
1 2 3 4
3 4 5 6
1
2
3
4
5
```

OUTPUT

```
Enter the first set elements: Enter the second set elements:
Menu
1. Union
2. Intersection
3. Difference (A - B)
4. Symmetric Difference
5. Exit
Enter your choice: Union: {1, 2, 3, 4, 5, 6}

Menu
1. Union
2. Intersection
3. Difference (A - B)
4. Symmetric Difference
5. Exit
Enter your choice: Intersection: {3, 4}

Menu
1. Union
2. Intersection
3. Difference (A - B)
4. Symmetric Difference
5. Exit
Enter your choice: Difference: {1, 2}

Menu
1. Union
2. Intersection
3. Difference (A - B)
4. Symmetric Difference
5. Exit
Enter your choice: Symmetric Difference: {1, 2, 5, 6}

Menu
1. Union
2. Intersection
3. Difference (A - B)
4. Symmetric Difference
```

5. Exit
Enter your choice: Program terminated.

Q20. Student Database

CODE

```
def main():  
    students = set()  
    while True:  
        choice = int(input("Enter your choice: "))  
        if choice == 1:  
            name = input("Enter student name: ")  
            students.add(name)  
            print("Student added successfully.")  
        else:  
            break  
  
main()
```

INPUT

```
1  
Ravi  
1  
Priya  
4  
6
```

OUTPUT

```
Enter your choice: Enter student name: Student added successfully.  
Enter your choice: Enter student name: Student added successfully.  
Enter your choice:
```